5 **(a)** Distinguish between electromotive force (e.m.f.) and potential difference (p.d.) using energy considerations.

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……..……..

……..……..

……..…… [2]

(b) In order to determine the internal resistance *r* of a 12.00 V battery, a student sets up the circuit shown in Fig. 5.1. The variable resistor CD has a maximum resistance of 1000 Ω.

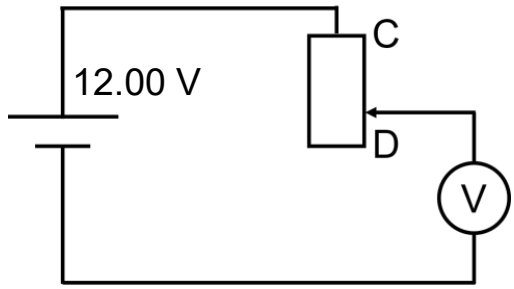

Fig. 5.1

It is found that when the sliding contact of the variable resistor is placed at C, the digital voltmeter reading is 11.99 V. When the sliding contact is moved down from C to D, the digital voltmeter reading drops from 11.99 V to 11.00 V.

(i) State the potential difference across the internal resistance of the battery when the sliding contact of the variable resistor is placed at C.

potential difference = …… V [1]

(ii) Explain why the presence of internal resistance in the battery reduces its output power.

……

……

…… [1]

(c) **(i)** Using the results of **(b)(i)**, show that $R = 1199\ r$ where R is the resistance of the voltmeter and r is the internal resistance.

[1]

(ii) Hence or otherwise, determine R and r.

R = …… Ω

r = …… Ω [3]